

Lab 2 Z-transfer Functions, Difference Equations, and Filter Implementation Using MATLAB

OBJECTIVES

1. To learn how to determine the difference equation given IIR (infinite impulse response) system coefficients.
2. To learn how to determine the IIR transfer function based on the given difference equation, and learn how to calculate and display frequency responses of the IIR systems and perform digital filtering.

PROCEDURE:

Part B: IIR Filters

An IIR system is given below

Numerator coefficients: $b=[0.5 \ 0.5]$

Denominator coefficients: $a=[1 \ -0.25]$;

- a. Determine its transfer function $H(z)=$
- b. Determine its difference equation $y(n)=$
- c. Given sampled values: $\text{sample}=[1:1:10]$, write a MATLAB program to perform the digital filtering

```
-----
sample = [1:1:10]
x = [0 sample]
y = [0 zeros(1,length(sample))]
for n = 2:1:length(x)
    y(n) = completed on your own
end
y(2:1:length(x))
-----
```

Using the standard MATLAB routine “filter()” for verification

```
yy = filter([0.5 0.5],[1 -0.25],sample)
```

Is the vector containing $y(2:1:\text{length}(\text{sample}))$ same as the vector yy ? _____

Two IIR filter coefficients are given below, respectively,

IIR system 1:

Numerator coefficients: $b_{BP} = [0.0201 \ 0.0000 \ -0.0402 \ 0.0000 \ 0.0201]$

Denominator coefficients: $a_{BP} = [1.0000 \ -2.1192 \ 2.6952 \ -1.6924 \ 0.6414]$

IIR system 2:

Numerator coefficients: $b_{HP} = [0.0938 \ -0.3753 \ 0.5630 \ -0.3753 \ 0.0938]$

Denominator coefficients: $a_{HP} = [1.0000 \ 0.0784 \ 0.8267 \ 0.2597 \ 0.1960]$

1. Determine the transfer function using z-transform for the first IIR system.
 2. Determine the difference equation for the first IIR system.
- Plot the frequency responses using the MATLAB function `freqz()` for each of z-transfer function.
- ```
MATLAB>>freqz(bBP,aBP,512,8000);axis([0 4000 -40 1]);
```
- Label and print your graph.

3. Perform similar operations for IIR systems with coefficients: aHP, and bHP.

Generate the sum of sinusoids for 1 second achieved from the Lab 1 for the duration of 1 second:

$$x(t) = 5 \cos(2\pi(500)t) + 5 \cos(2\pi(1200)t + 0.25\pi) + 5 \cos(2\pi(1800)t + 0.5\pi)$$

using a sampling rate of 8 KHz.

5. Write MATLAB programs using the MATLAB function **filter()** according to two IIR systems.
- Perform digital filtering using each of the developed programs, and plot the input signal spectrum and the output signal spectrum using the frequency resolution of 1 Hz, respectively. Print and label spectrum.
  - Listen to and compare the input signal and output signal from each filter, respectively.

Discuss the obtained results.

What frequency components/component (frequency locations of the spectral peak(s)) are/is shown in the output spectrum?

### OPTIONAL BUT ATTRACTS EXTRA MARKS

6. Use speech data with a sampling rate of 8000 Hz provided by your instructor,
- ```
MATLAB>>load we.dat;
MATLAB>>x=10*we/max(abs(we)); %scale the speech data
```
- Perform digital filtering using each of the developed programs, and calculate and plot the input signal spectrum and the output spectrum with a frequency resolution of 4 Hz, respectively. Print and label the spectrum.
 - Listen to and compare the input signal and output signal from each filter, respectively.
- Discuss the obtained results.

Organize your design, plots and program lists.